

(iii) $2W = 37 \sin p$

B1

[1]

- 3 (a) sum of (random) kinetic and potential energies
of the atoms/molecules of the substance

M1

A1

[2]

- (b) (i) potential energy unchanged as atoms remain in same positions
allow 'reduced because atoms slightly closer together'

M1

vibrational kinetic energy reduced because temperature lower
so internal energy less

M1

A1

[3]

- (ii) potential energy increases because separation increases
kinetic energy unchanged because temperature unchanged
so internal energy increases

M1

M1

A1

[3]

- 4 (a) mass per unit volume (*ratio idea must be clear not units*)

B1

[1]